

Lecture 17: Constructions by Straightedge and Compass

MAT205 Abstract Algebra II

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Part I: Constructible Numbers and the Field F

Three Problems from Antiquity

The instruments. A *straightedge* draws the line through two known points (no marks, no scale); a *compass* draws the circle centered at a known point through another known point. We start from a segment declared to have length 1.

Three constructions the Greeks could not perform, each now read as “*construct a segment of this length*”:

- **Doubling the cube** (the Delian problem): the edge of a cube of twice the unit volume — length $\sqrt[3]{2}$.
- **Trisecting the angle**: a third of a given angle — already 60° resists, i.e. one cannot reach $\cos 20^\circ$.
- **Squaring the circle**: a square equal in area to the unit circle — side $\sqrt{\pi}$.

From the 4th century BC, Greek geometers solved all three with other tools (conic sections, the marked-ruler *neusis*), never with straightedge and compass, and could not prove it impossible. Nineteenth-century field theory settles all three.

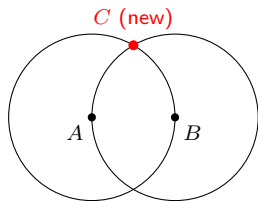
The Rules of the Game, Made Precise

Constructed points are built recursively from $(0,0)$ and $(1,0)$:

- draw the *line* through two constructed points;
- draw the *circle* centered at a constructed point through a constructed point;
- a new point is an intersection of two such loci.

A real number α is **constructible** if some constructed segment has length $|\alpha|$; a point (x,y) is constructible iff x and y are.

Classical Euclidean moves are available: erect/drop perpendiculars, copy a length, draw parallels, bisect segments and angles. From the unit segment one locates every point with rational coordinates.



two unit circles centered at $A = (0,0)$ and $B = (1,0)$ meet at the new point

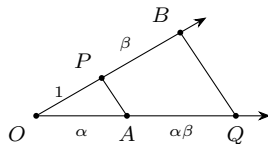
$$C = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \text{ (Euclid I.1)}$$

Constructible Numbers Form a Field

Theorem. If α, β are constructible, then so are $\alpha + \beta$, $\alpha - \beta$, $\alpha\beta$, and α/β (for $\beta \neq 0$).

Proof.

- *Sum/difference:* lay the length $|\beta|$ off along the line carrying $|\alpha|$ with the compass.
- *Product:* on one ray from O lay $|OP| = 1$ then $|PB| = |\beta|$; on another lay $|OA| = |\alpha|$. The parallel to PA through B meets ray OA at Q , and by the intercept theorem $\frac{|OP|}{|PB|} = \frac{|OA|}{|AQ|}$, so $|AQ| = |\alpha\beta|$.
- *Quotient:* the analogous parallel construction yields $|\alpha/\beta|$. ■



product by similar triangles: $PA \parallel BQ$

Corollary. $F := \{\text{constructible reals}\}$ is a subfield of $\mathbb{R}$, and $\mathbb{Q} \subseteq F$.

Square Roots Are Constructible

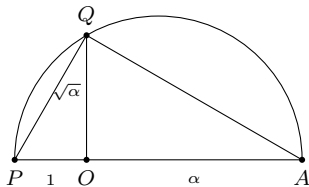
Theorem. If $\alpha > 0$ is constructible, then $\sqrt{\alpha}$ is constructible.

Proof (geometric mean). On a line mark P, O, A with $|PO| = 1$ and $|OA| = \alpha$. Draw the semicircle on diameter PA and the perpendicular to PA at O , meeting the semicircle at Q . By Thales' theorem the angle $\angle PQA$ is right, so triangles OPQ and OQA are similar:

$$\frac{|OQ|}{|OP|} = \frac{|OA|}{|OQ|} \implies |OQ|^2 = 1 \cdot \alpha = \alpha,$$

hence $|OQ| = \sqrt{\alpha}$. ■

So F is a field that is also closed under $\sqrt{}$ of positive elements — the one operation a compass adds beyond arithmetic.



Each Step Adjoins at Most One Square Root

Suppose every point built so far has coordinates in a subfield $H \subseteq \mathbb{R}$ (start $H = \mathbb{Q}$).
Then:

- lines through two such points: $ax + by + c = 0$ with $a, b, c \in H$;
- circles centered at such a point through such a point: $x^2 + y^2 + dx + ey + f = 0$ with $d, e, f \in H$.

A new point is an intersection of two of these:

- **line** $\cap$ **line**: two linear equations over H — the solution stays in H , no new number;
- **circle** $\cap$ **circle**: subtracting the equations cancels $x^2 + y^2$, leaving a line over H (the *radical axis*); this reduces to the next case;
- **line** $\cap$ **circle**: substitution gives a quadratic over H , whose roots lie in $H(\sqrt{d})$ for some $d \in H$, $d \geq 0$.

So each construction step keeps the coordinates inside H or a quadratic extension $H(\sqrt{d})$: the degree over the current field is 1 or 2.

The Tower Characterization of F

Theorem. For $\alpha \in \mathbb{R}$, the following are equivalent.

1. α is constructible.
2. There is a tower $\mathbb{Q} = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n$ with $\alpha \in K_n$ and each $[K_i : K_{i-1}] = 2$ (so $K_i = K_{i-1}(\sqrt{d_i})$, $d_i \in K_{i-1}$, $d_i > 0$).
3. α is obtained from $\mathbb{Q}$ by finitely many field operations and square roots of positive numbers.

Proof.

- $(1) \Rightarrow (2)$: each new point adjoins one square root; collect them into a tower whose top field K_n contains α .
- $(2) \Rightarrow (3)$: each step $K_{i-1}(\sqrt{d_i})$ is field operations plus one square root.
- $(3) \Rightarrow (1)$: F is a field containing $\mathbb{Q}$ and closed under positive square roots, so it contains every such expression. ■

Part II: The Degree Criterion

Constructible Numbers Have 2-Power Degree

Corollary. If $\alpha \in \mathbb{R}$ is constructible, then $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^r$ for some integer $r \geq 0$.

Proof. By the tower characterization $\alpha \in K_n$ with each $[K_i : K_{i-1}] = 2$, so by the tower law

$$[K_n : \mathbb{Q}] = \prod_{i=1}^n [K_i : K_{i-1}] = 2^n.$$

Since $\mathbb{Q} \subseteq \mathbb{Q}(\alpha) \subseteq K_n$, the tower law gives $[K_n : \mathbb{Q}] = [K_n : \mathbb{Q}(\alpha)] [\mathbb{Q}(\alpha) : \mathbb{Q}]$, so $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides 2^n and is a power of 2. ■

This criterion is due to Pierre Wantzel (1837). **Use:** to show a length is *not* constructible, exhibit a number whose degree over $\mathbb{Q}$ is not a power of 2 — most cheaply, degree 3.

Necessary, Not Sufficient

The converse of the degree criterion is **false**.

There exist α with $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$ that are not constructible: a root of an irreducible quartic whose splitting field has Galois group S_4 has degree 4, yet cannot be reached by a tower of quadratic extensions.

The sharp condition is Galois-theoretic: α is constructible exactly when the *Galois closure* of $\mathbb{Q}(\alpha)/\mathbb{Q}$ — the smallest Galois extension of $\mathbb{Q}$ containing α — is a 2-group. (Over $\mathbb{Q}$ every algebraic extension is separable, so the Galois closure coincides with the normal closure of L15 and equals the splitting field of $\text{irr}(\alpha, \mathbb{Q})$.) We prove this once the Galois correspondence is in hand.

For the three classical problems the necessary direction already suffices, so we settle them first.

Part III: The Three Impossibility Proofs

Doubling the Cube Is Impossible

Theorem. One cannot construct, with straightedge and compass, the edge of a cube of twice the volume of a given cube.

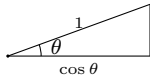
Proof. Take the given edge as the unit; the target edge has length $\sqrt[3]{2}$. The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ (Eisenstein at $p = 2$), so

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3.$$

Since 3 is not a power of 2, the degree criterion shows $\sqrt[3]{2}$ is not constructible. ■

Trisecting the Angle: Reducing to a Cubic

Lemma (angle $\leftrightarrow$ cosine). An angle θ is constructible iff a segment of length $|\cos \theta|$ is constructible. (*With the unit segment as hypotenuse, the adjacent leg has length $|\cos \theta|$.*)



Triple-angle identity. From $\cos 3\theta = \cos(2\theta + \theta)$ and the addition formulas,

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

Put $\theta = 20^\circ$, so $\cos 3\theta = \cos 60^\circ = \frac{1}{2}$. With $\alpha = \cos 20^\circ$,

$$4\alpha^3 - 3\alpha = \frac{1}{2}, \quad \text{i.e.} \quad 8\alpha^3 - 6\alpha - 1 = 0.$$

Trisecting the Angle Is Impossible

Theorem. The angle 60° cannot be trisected by straightedge and compass.

Proof. 60° is constructible (equilateral triangle). Trisecting it means constructing $\alpha = \cos 20^\circ$, a root of $f(x) = 8x^3 - 6x - 1$. Substituting $x = y + \frac{1}{2}$,

$$f\left(y + \frac{1}{2}\right) = 8y^3 + 12y^2 - 3,$$

which is Eisenstein at $p = 3$ ($3 \nmid 8$, $3 \mid 12$, $3 \mid -3$, $9 \nmid -3$), hence irreducible over $\mathbb{Q}$. So f is irreducible too, and

$$[\mathbb{Q}(\cos 20^\circ) : \mathbb{Q}] = 3.$$

As 3 is not a power of 2, $\cos 20^\circ$ is not constructible, so 60° is not trisectable. ■

Some angles do trisect (e.g. $90^\circ \rightarrow 30^\circ$); the theorem is that no single method trisects every angle.

Squaring the Circle Is Impossible

Theorem. One cannot construct a square equal in area to a given circle.

Cited input (Lindemann, 1882). π is *transcendental* over $\mathbb{Q}$: it is a root of no nonzero polynomial in $\mathbb{Q}[x]$. (The proof is beyond this course; we use it as a black box.)

Proof. For the unit circle (area π) the square has side $\sqrt{\pi}$. If $\sqrt{\pi}$ were constructible then so would $\pi = (\sqrt{\pi})^2$ be, hence π would be algebraic of degree a power of 2. But a transcendental number lies in no finite extension of $\mathbb{Q}$ — contradiction. ■



Ferdinand von Lindemann
(1852–1939): proved π
transcendental in 1882.

Part IV: Regular Polygons and the Galois Criterion

Constructibility in the Plane

Identify the plane with $\mathbb{C}$. A complex number $z = a + bi$ is **constructible** iff $a, b \in F$; the constructible complex numbers form a subfield $\mathbb{K} \subseteq \mathbb{C}$ containing $\mathbb{Q}$.

Closed under square roots. If $z = \rho e^{i\psi}$ is constructible, one constructs $\sqrt{\rho}$ (geometric mean) and bisects the angle ψ , giving $\sqrt{\rho} e^{i\psi/2}$. So $\mathbb{K}$ is closed under $\sqrt{}$.

Complex tower characterization. For $z \in \mathbb{C}$, z is constructible iff z lies in a tower $\mathbb{Q} = K_0 \subseteq \cdots \subseteq K_n$ of subfields of $\mathbb{C}$ with each $[K_i : K_{i-1}] = 2$. (Each step adjoins a $\sqrt{}$, which $\mathbb{K}$ contains; conversely $z = a + bi$ with $a, b \in F$ sits in a real quadratic tower, and $i = \sqrt{-1}$ is one more quadratic step.)

The Regular n -gon Reduces to $\cos(2\pi/n)$

Inscribe the regular n -gon in the unit circle with one vertex at 1. Its vertices are the powers ζ_n^k of

$$\zeta_n = e^{2\pi i/n} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}.$$

Therefore:

- the n -gon is constructible $\iff \zeta_n$ is constructible;
- ζ_n is constructible $\iff \cos(2\pi/n)$ is constructible, since then $\sin(2\pi/n) = \sqrt{1 - \cos^2(2\pi/n)}$ is constructible too.

So the geometric question becomes: *for which n is $\cos(2\pi/n)$ constructible?* The Galois criterion answers it.

The Galois Criterion for Constructibility

Theorem. Let $\alpha \in \mathbb{C}$ be algebraic over $\mathbb{Q}$, and let L be the Galois closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$ (the splitting field of $\text{irr}(\alpha, \mathbb{Q})$). Then

$$\alpha \text{ constructible} \iff [L : \mathbb{Q}] = 2^m \iff \text{Gal}(L/\mathbb{Q}) \text{ is a 2-group.}$$

Proof of $\Leftarrow$ (sufficiency). Let $G = \text{Gal}(L/\mathbb{Q})$ have order 2^m . A finite 2-group has a chain of subgroups

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_m = \{1\}, \quad [G_i : G_{i+1}] = 2.$$

By the Galois correspondence this gives a tower of fields

$$\mathbb{Q} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_m = L, \quad [F_{i+1} : F_i] = [G_i : G_{i+1}] = 2.$$

Each step is a quadratic extension, so every element of L — in particular α — lies in a quadratic tower and is constructible.

The Galois Criterion: Necessity

Proof of $\Rightarrow$. If α is constructible it lies in a quadratic tower $\mathbb{Q} = K_0 \subseteq \cdots \subseteq K_n$. For each $\mathbb{Q}$ -embedding $\sigma : K_n \hookrightarrow \mathbb{C}$, the image σK_n is again a quadratic tower $(\sigma K_i = \sigma K_{i-1}(\sqrt{\sigma d_i}))$.

Let M be the compositum of all these conjugate towers. Adjoining the finitely many square roots $\sqrt{\sigma d_i}$ one at a time builds M by steps of degree 1 or 2, so $[M : \mathbb{Q}] = 2^k$. Now M contains every conjugate $\sigma\alpha$, hence contains the splitting field $L \subseteq M$. By the tower law $[L : \mathbb{Q}]$ divides 2^k , so it is a power of 2; and as $L/\mathbb{Q}$ is Galois, $|\text{Gal}(L/\mathbb{Q})| = [L : \mathbb{Q}] = 2^m$. ■

This recovers the degree criterion ($[\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides 2^m) and explains its failure as a converse: the *closure* must be a 2-group, not merely $\mathbb{Q}(\alpha)$ of 2-power degree.

Cyclotomic Degree and Galois Group

The n -gon field. The minimal polynomial of ζ_n over $\mathbb{Q}$ is the cyclotomic polynomial Φ_n , irreducible of degree $\varphi(n)$ (Euler's totient). Hence

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n), \quad \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

via $\sigma_a : \zeta_n \mapsto \zeta_n^a$ for $\gcd(a, n) = 1$. (The irreducibility of Φ_n is a classical theorem of Gauss; the group is abelian of order $\varphi(n)$.)

Consequence. $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is Galois and abelian, so it is its own Galois closure. By the Galois criterion,

$$\zeta_n \text{ constructible} \iff \varphi(n) \text{ is a power of } 2.$$

The Gauss–Wantzel Theorem

Fermat prime. A prime of the form $F_t = 2^{2^t} + 1$; the known ones are 3, 5, 17, 257, 65537.

Theorem. The regular n -gon ($n \geq 3$) is constructible iff

$$n = 2^k p_1 p_2 \cdots p_s,$$

with $k \geq 0$ and the p_i distinct Fermat primes.

Proof. The n -gon is constructible iff $\varphi(n)$ is a power of 2. For $n = 2^{a_0} p_1^{a_1} \cdots p_s^{a_s}$,

$$\varphi(n) = \varphi(2^{a_0}) \prod_i p_i^{a_i-1} (p_i - 1),$$

and $\varphi(2^{a_0})$ is always a power of 2. So $\varphi(n)$ is a power of 2 iff every $a_i = 1$ and every $p_i - 1$ is a power of 2; and $p - 1 = 2^e$ forces $e = 2^t$ (an odd factor $v > 1$ of $e = uv$ would give $2^u + 1 \mid 2^e + 1 = p$), i.e. p a Fermat prime. ■

Constructible: 3, 4, 5, 6, 8, 10, 12, 15, 16, 17, ... Not: 7, 9, 11, 13, 14, ... ($\varphi(7) = \varphi(9) = 6$).

Gauss and the Regular 17-gon

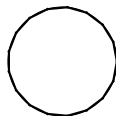
In 1796, aged 19, Gauss found the regular 17-**gon** constructible — $\varphi(17) = 16 = 2^4$ — and opened his *Disquisitiones Arithmeticae* (1801) with it. He gave the explicit value

$$\begin{aligned}\cos \frac{2\pi}{17} = & -\frac{1}{16} + \frac{1}{16}\sqrt{17} + \frac{1}{16}\sqrt{34 - 2\sqrt{17}} \\ & + \frac{1}{8}\sqrt{17 + 3\sqrt{17} - \sqrt{34 - 2\sqrt{17}} - 2\sqrt{34 + 2\sqrt{17}}},\end{aligned}$$

a tower of nested square roots — exactly a quadratic tower. Gauss stated the full classification and proved the constructible half.



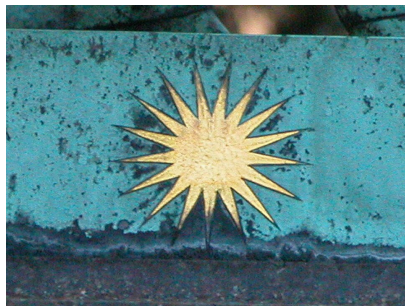
C. F. Gauss (1777–1855).



Gauss's Tomb and Memorial



Gauss's grave, Albani-Friedhof, Göttingen.



17-pointed star on the Gauss-Denkmal
(Braunschweig; statue 1880).

Gauss reportedly asked for a regular 17-gon on his tombstone; the mason refused, saying it would look like a circle. The 17-pointed star on the Brunswick memorial stands in.